

eCRF DESIGN SPECIFICATION

Study Information

Study Title:

A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Study to Evaluate the Efficacy and Safety of XYZ-101 in Patients with Type 2 Diabetes Mellitus

Study Population: Adult patients with Type 2 Diabetes Mellitus

Planned Number of Subjects: 120

Study Phase: Phase 3

Study Design: Randomized, Double-Blind, Placebo-Controlled Clinical Trial

Investigational Product: XYZ101

Comparator: Placebo

1. Purpose

The purpose of this eCRF Design Specification is to define the electronic Case Report Form (eCRF) structure and data fields required for the collection of clinical trial data for the XYZ101 Phase III clinical trial.

The eCRF is designed based on the study protocol, Data Management Plan (DMP), and Data Requirement Matrix. CDASH principles are applied to promote standardized and consistent collection of clinical data.

2. Primary Objective

To evaluate the efficacy of XYZ101 compared with placebo in improving glycemic control in patients with Type 2 diabetes mellitus.

3. Primary Efficacy Endpoint

The primary efficacy endpoint is the change in HbA1c from baseline to the protocol-specified post-baseline assessment.

4. eCRF Design Principles

The eCRF will be designed to:

- Capture all protocol-required clinical data.
- Support the primary and secondary study objectives and endpoints.
- Collect data according to the scheduled study visits and assessments.
- Apply CDASH principles where applicable.
- Use standardized terminology and appropriate data types.
- Minimize unnecessary or duplicate data collection.
- Support automated edit checks and data validation.
- Facilitate downstream data cleaning, review, and analysis.
- Maintain appropriate blinding of treatment assignment.

5. eCRF Structure

The following eCRF forms will be included in the study database based on the protocol and data requirements:

Form No.	eCRF Form Name
01	Subject Identification & Demographics
02	Informed Consent
03	Eligibility – Inclusion & Exclusion Criteria
04	Medical History
05	Prior & Concomitant Medications
06	Vital Signs
07	Laboratory Assessments
08	Efficacy Assessments
09	Adverse Events
10	Study Drug Exposure & Compliance
11	Study Completion / Early Termination

6. eCRF Field Specification

Each eCRF form will contain fields defined according to the study data requirements.

The following attributes will be specified for each field:

- Field Name
- Field Label
- Data Type
- Required/Optional
- CDASH Concept

- Controlled Terminology, where applicable
- Permitted Values, where applicable
- Edit Check
- Comments/Notes, where applicable

7. eCRF Form 01 – Subject Identification & Demographics

7.1 Purpose

To capture the subject identification and demographic information required for study participation and data management.

7.2 Field Specification

Field Name	Field Label	Data Type	Required	CDASH Concept	Edit Check
SITEID	Site ID	Text	Yes	Site Identifier	Must correspond to a valid study site
SCRNNO	Screening Number	Text	Yes	Screening Identifier	Must be unique within the site
SUBJID	Subject ID	Text	Yes	Subject Identifier	Must correspond to a valid subject
BRTHDTC	Date of Birth	Date	Yes	Date of Birth	Cannot be a future date
AGE	Age	Numeric	Yes	Age	Must meet protocol-defined age criteria
SEX	Sex	Codelist	Yes	Sex	Must be a permitted value
RACE	Race	Codelist	Yes	Race	Must be a permitted value
ETHNIC	Ethnicity	Codelist	Yes	Ethnicity	Must be a permitted value

8. eCRF Form 02 – Informed Consent

8.1 Purpose

To document the subject's informed consent before participation in study-specific procedures.

8.2 Field Specification

Field Name	Field Label	Data Type	Required	CDASH Concept	Edit Check
ICFYN	Informed Consent Obtained	Codelist	Yes	Informed Consent	Must be Yes before study procedures
ICFDTC	Informed Consent Date	Date	Yes	Informed Consent Date	Cannot be after screening date
ICFVER	Informed Consent Form Version	Text	Yes	Consent Form Version	Must be a valid approved version

9. eCRF Form 03 – Eligibility: Inclusion & Exclusion Criteria

9.1 Purpose

To document assessment of the protocol-defined inclusion and exclusion criteria before subject enrollment.

9.2 Field Specification

The form will capture each protocol-defined eligibility criterion as a separate assessment.

Field Name	Field Label	Data Type	Required	CDASH Concept	Edit Check
INCLYN	Inclusion Criteria Met	Yes/No	Yes	Inclusion Criteria Assessment	All required inclusion criteria must be satisfied
EXCLYN	Exclusion Criteria Met	Yes/No	Yes	Exclusion Criteria Assessment	No exclusion criterion should be met
ELIGYN	Eligibility for Study	Yes/No	Yes	Eligibility Status	Must be consistent with individual criteria
ELIGDTC	Eligibility Date	Date	Yes	Eligibility Date	Must occur after informed consent

10. eCRF Form 04 – Medical History

10.1 Purpose

To capture relevant medical conditions and medical history prior to study treatment.

10.2 Field Specification

Field Name	Field Label	Data Type	Required	CDASH Concept	Edit Check
MHYN	Medical History Present	Yes/No	Yes	Medical History	If No, no medical history records should be entered
MHTERM	Medical History Term	Text	Conditional	Medical History Term	Required when MHYN = Yes
MHSTDTC	Start Date	Date	Conditional	Start Date	Cannot be after end date
MHONGO	Ongoing	Yes/No	Conditional	Ongoing Indicator	Required when applicable

11. eCRF Form 05 – Prior & Concomitant Medications

11.1 Purpose

To capture medications taken prior to and during the study.

11.2 Field Specification

Field Name	Field Label	Data Type	Required	CDASH Concept	Edit Check
CMTRT	Medication Name	Text	Yes	Medication	Required for each medication
CMSTDTC	Start Date	Date	Yes	Start Date	Cannot be after stop date
CMENDTC	Stop Date	Date	Conditional	Stop Date	Must be $\geq$ start date
CMDOS	Dose	Numeric/Text	Yes	Dose	Must be a valid dose
CMROUTE	Route	Codelist	Yes	Route	Must be a permitted value

CMONGO	Ongoing	Yes/No	Yes	Ongoing Indicator	Consistent with stop date
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12. eCRF Form 06 – Vital Signs

12.1 Purpose

To capture protocol-specified vital sign measurements at applicable study visits.

12.2 Field Specification

Field Name	Field Label	Data Type	Required	CDASH Concept	Edit Check
VSDTC	Assessment Date	Date	Yes	Vital Signs Date	Must correspond to study visit
SYSBP	Systolic Blood Pressure	Numeric	Yes	Systolic Blood Pressure	Must be within defined validation range
DIABP	Diastolic Blood Pressure	Numeric	Yes	Diastolic Blood Pressure	Must be within defined validation range
PULSE	Pulse Rate	Numeric	Yes	Pulse Rate	Must be within defined validation range
TEMP	Temperature	Numeric	Yes	Temperature	Must be within defined validation range
WEIGHT	Weight	Numeric	Yes	Weight	Must be within defined validation range

13. eCRF Form 07 – Laboratory Assessments

13.1 Purpose

To capture protocol-specified laboratory assessments for efficacy and safety evaluation.

13.2 Field Specification

Field Name	Field Label	Data Type	Required	CDASH Concept	Edit Check
LBTEST	Lab Test	Codelist/Text	Yes	Laboratory Test	Must be a protocol-specified test
LBDTC	Collection Date	Date	Yes	Specimen Collection Date	Must correspond to study visit
LBORRES	Lab Result	Numeric/Text	Yes	Laboratory Result	Must be a valid result
LBORRESU	Result Unit	Text/Codelist	Yes	Unit	Must correspond to test
LBORNR	Reference Range	Numeric/Text	Conditional	Reference Range	Must be appropriate for test

14. eCRF Form 08 – Efficacy Assessments

14.1 Purpose

To capture efficacy data required to evaluate the effect of XYZ101 compared with placebo on glycemic control.

14.2 Primary Efficacy Assessment

HbA1c will be collected at baseline and at protocol-specified post-baseline assessment timepoints.

Field Name	Field Label	Data Type	Required	CDASH Concept	Edit Check
EFDTC	Assessment Date	Date	Yes	Assessment Date	Must correspond to scheduled visit
VISIT	Visit	Codelist	Yes	Visit	Must be a valid study visit
LBHBA1C	HbA1c Result	Numeric	Yes	HbA1c Result	Must be within defined validation range
LBHBA1CU	HbA1c Unit	Codelist	Yes	Unit	Must be %

EFSTAT	Assessment Status	Codelist	Yes	Assessment Status	Must be a permitted value
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15. eCRF Form 09 – Adverse Events

15.1 Purpose

To capture adverse events occurring during the study.

15.2 Field Specification

Field Name	Field Label	Data Type	Required	CDASH Concept	Edit Check
AETERM	Adverse Event Term	Text	Yes	Adverse Event	Required for each AE
AESTDTC	Start Date	Date	Yes	Start Date	Cannot be after end date
AEENDTC	End Date	Date	Conditional	End Date	Must be $\geq$ start date
AESEV	Severity	Codelist	Yes	Severity	Must be a permitted value
AESER	Serious	Yes/No	Yes	Seriousness	Must be a permitted value
AEREL	Relationship to Study Treatment	Codelist	Yes	Causality	Must be a permitted value
AEACT	Action Taken	Codelist	Yes	Action Taken	Must be a permitted value
AEOUT	Outcome	Codelist	Yes	Outcome	Must be a permitted value

16. eCRF Form 10 – Study Drug Exposure & Compliance

16.1 Purpose

To capture information related to study drug administration, exposure, and compliance while maintaining the study blind.

The eCRF will not display the actual treatment identity (XYZ101 or placebo) to blinded study personnel where such information is intended to remain masked.

Field Name	Field Label	Data Type	Required	CDASH Concept	Edit Check
EXDTC	Study Drug Date	Date	Yes	Study Drug Date	Must correspond to study period
EXDOSE	Dose Administered	Numeric/Text	Yes	Dose	Must be protocol-permitted
EXDOSU	Dose Unit	Codelist	Yes	Dose Unit	Must be valid
EXSTAT	Administration Status	Codelist	Yes	Administration Status	Must be a permitted value
EXCOMP	Compliance Assessment	Numeric/Percentage	Conditional	Drug Compliance	Must be within valid range

17. eCRF Form 11 – Study Completion / Early Termination

17.1 Purpose

To document study completion or early discontinuation of the subject.

Field Name	Field Label	Data Type	Required	CDASH Concept	Edit Check
DSSTAT	Study Completion Status	Codelist	Yes	Completion Status	Must be a permitted value
DSDTC	Completion Date	Date	Conditional	Discontinuation Reason	Must be consistent with study timeline
DSREAS	Discontinuation Reason	Codelist	Conditional	Discontinuation Reason	Required if subject discontinues
DSLVST	Last Study Visit	Codelist	Yes	Last Study Visit	Must be a valid study visit

18. CDASH Implementation

CDASH concepts will be considered during eCRF development to promote standardized data collection and facilitate downstream CDISC processes.

The eCRF fields will be reviewed for:

- Standard variable naming where applicable
- Standardized question/field concepts
- Appropriate data types
- Consistent use of controlled terminology
- Standardized date formats
- Consistent collection of clinical data
- Alignment with protocol requirements

The final CDASH mapping and SDTM mapping will be documented separately in the Annotated eCRF.

19. Blinding Considerations

Because the study is double-blind, treatment assignment will remain masked to blinded study personnel.

Randomization and treatment allocation will be managed through the appropriate randomization/IRT/RTSM process. The routine eCRF will capture only the treatment-related information required for study data collection without unnecessarily revealing the treatment identity.

20. eCRF Review and Approval

The eCRF design will be reviewed to confirm:

- Alignment with the study protocol
- Alignment with the DMP
- Completeness of required data collection
- Appropriate CDASH implementation
- Appropriate controlled terminology
- Appropriate edit checks
- Maintenance of study blinding
- Suitability for downstream data management and analysis

The finalized eCRF design will be approved according to the study-specific review and approval process before database implementation.